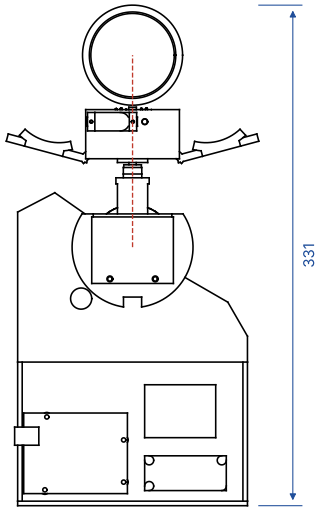
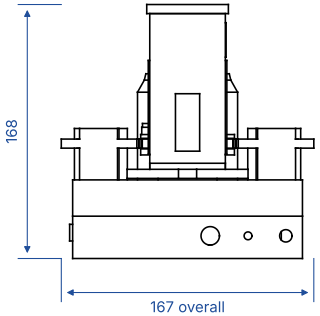
CA-000 Concept A assembly, HOME pose (arm at START)

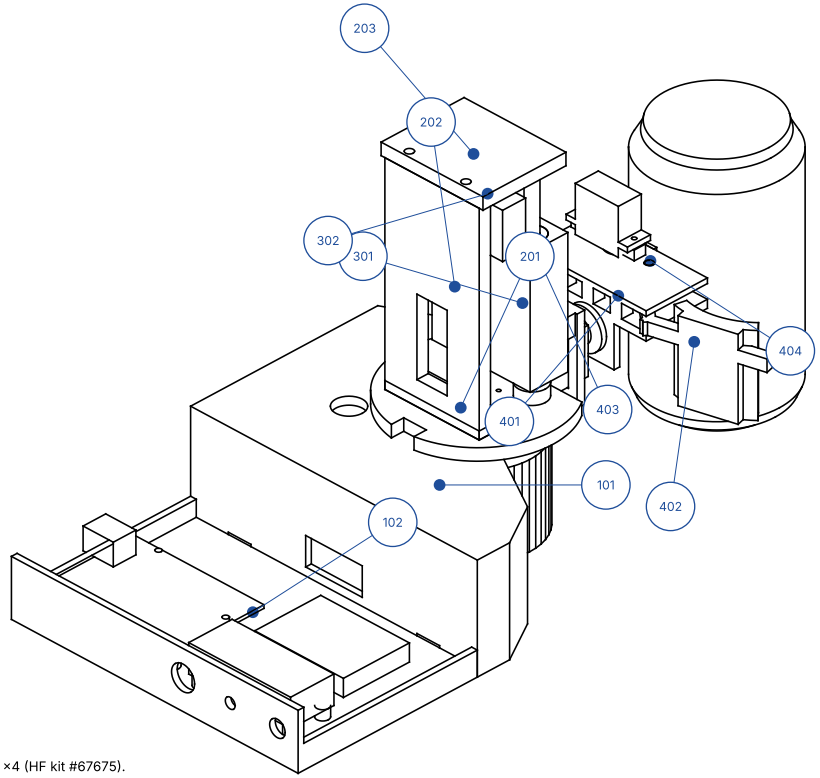
TOP VIEW



R 127 mm (5.0 in): turntable axis → can centre (dashed)



FRONT VIEW



ISOMETRIC (not to scale)

PARTS LIST (full BOM: docs/05_bom.md)

CA-101	Base	PETG	1
CA-102	Electronics tray	PETG	1
CA-201	Turret plate	PETG	1
CA-202	Spine	PETG	1
CA-203	Top cap	PETG	1
CA-301	Carriage + arm	PETG	1
CA-302	Lift link	PETG	1
CA-401	Gripper frame	PETG	1
CA-402	Finger, right	PETG	1
CA-403	Finger, left	PETG	1
CA-404	Can bumper	silicone	1
CA-P02	Servo MG996R (turntable)	buy	1
CA-P03	Servo MG90S (lift, wrist)	buy	2
CA-P04	Servo SG90 (gripper)	buy	1
CA-P05	Bearing 6807-2RS	buy	1
CA-P06	Linear bearing LM6UU	buy	4
CA-P07	Steel rod 6 × 112 mm	buy	2
CA-P01	Arduino Uno R3	buy	1
CA-P09	Buck converter XL4015	buy	1

Balloon numbers = part number without the CA- prefix. Fasteners: #6-32 × ½″ ×8, #8-32 × ½″ ×4 (HF kit #67675).
Motion ranges: turret 0 → −90°, lift 0 → 15.3 mm, wrist 0 → 180°, fingers 0 → 75°.

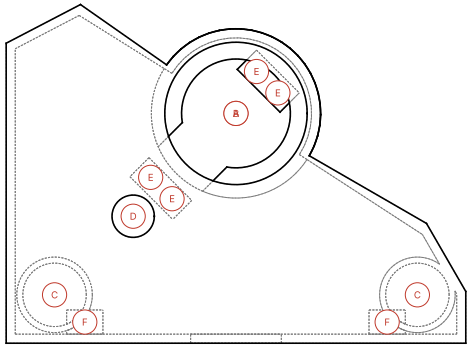
PART NO.	**CA-000**
TITLE	**Concept A assembly (HOME pose)**
MASS	≈ 571 g robot + 15 g can (CAD)
SCALE	1:5 (front, top)
UNITS	mm
SOURCE	cad/step/concept_a_assembly.step
PROJECTION	Third angle ⊕◁
SHEET	1 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

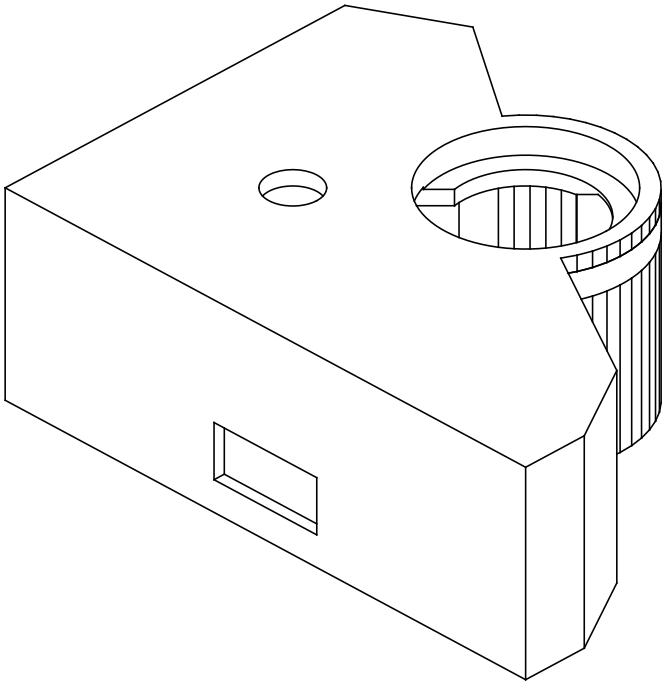
CA-101 Base

NOTES

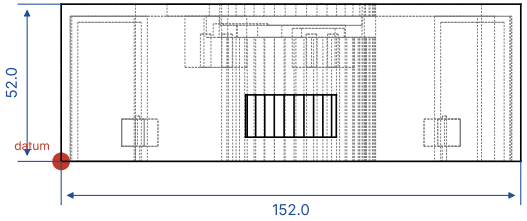
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

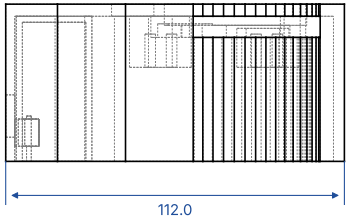




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	47.10	1	Z	7.0	(76.0, 76.0)	6807 outer race press fit
B	36.00	1	Z	4.0	(76.0, 76.0)	bearing seat floor bore
C	21.00	2	Z	46.0	(16.0, 16.0); (136.0, 16.0)	ballast washer pocket
D	14.00	1	Z	4.0	(42.0, 42.0)	cable hole from the turret
E	3.50	4	Z	11.0	(47.8, 54.9); (54.9, 47.8) +2	#8-32 pilot
F	2.90	2	Z	15.0	(26.0, 7.0); (126.0, 7.0)	#6-32 pilot (self-threading)

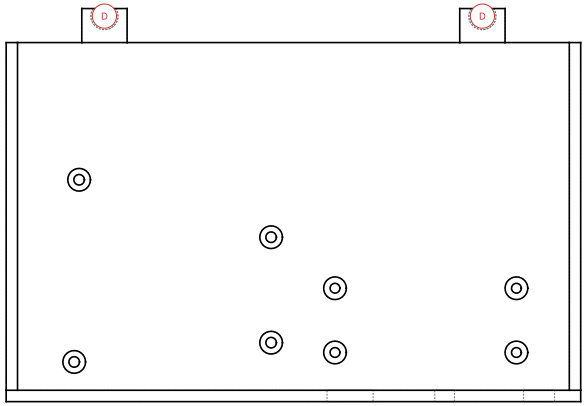
PART NO.	**CA-101**
TITLE	**Base**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	102.3 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	2:5
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/base.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	2 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

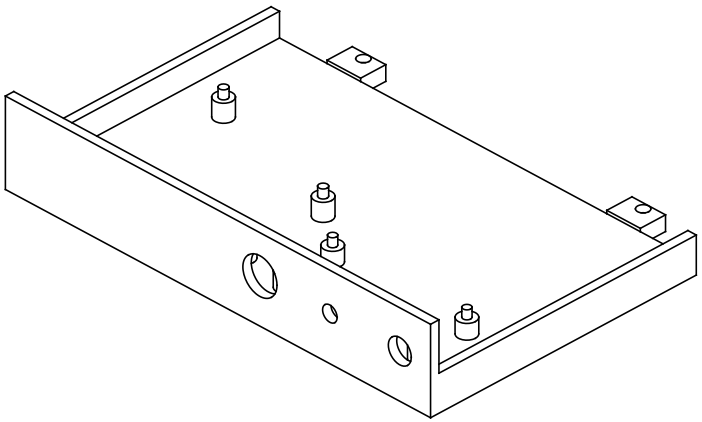
CA-102 Electronics tray

NOTES

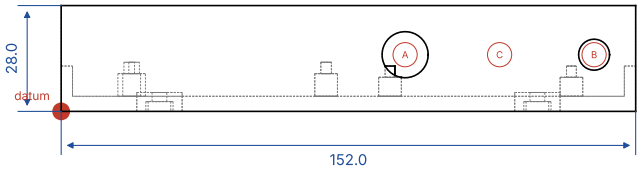
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

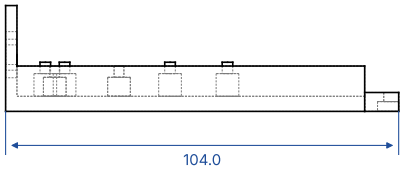




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	12.20	1	Y	3.0	(91.0, 15.0)	12 mm START button
B	8.20	1	Y	3.0	(141.0, 15.0)	DC panel jack
C	5.20	1	Y	3.0	(116.0, 15.0)	5 mm status LED
D	3.90	2	Z	2.4	(26.0, 102.0); (126.0, 102.0)	#6-32 clearance / running

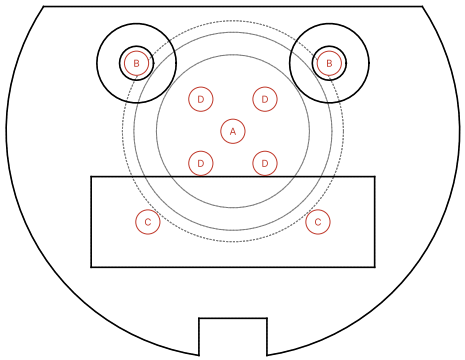
PART NO.	**CA-102**
TITLE	**Electronics tray**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	57.0 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	1:2
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/electronics_tray.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	3 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

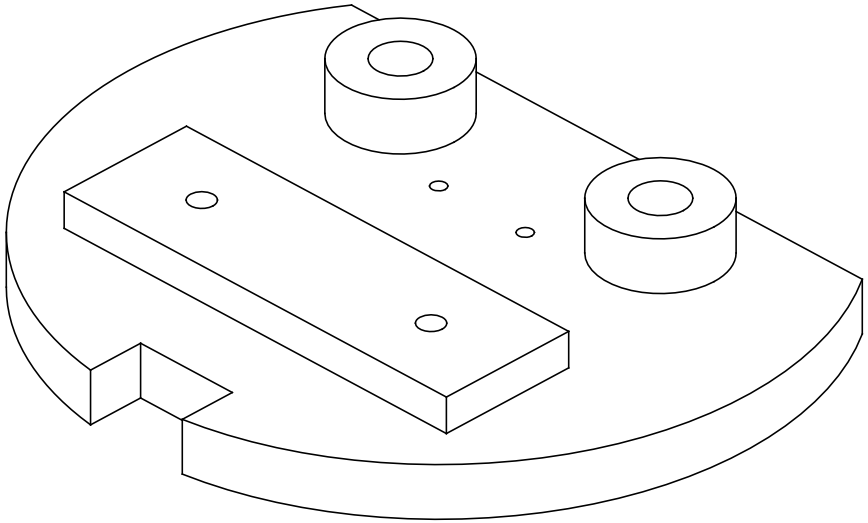
CA-201 Turret plate

NOTES

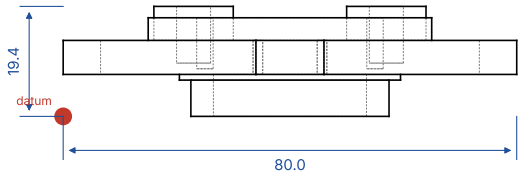
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

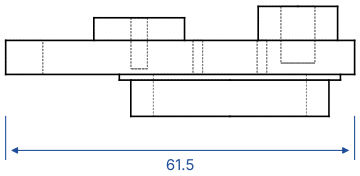




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	27.00	1	Z	7.4	(40.0, 39.5)	recess for the Ø25 MG996R horn
B	6.00	2	Z	10.0	(23.0, 51.5); (57.0, 51.5)	6 mm rod press fit
C	2.90	2	Z	9.0	(25.0, 23.5); (55.0, 23.5)	#6-32 pilot (self-threading)
D	1.70	4	Z	6.0	(34.3, 33.9); (34.3, 45.2) +2	servo-bag screw pilot

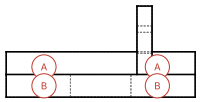
PART NO.	**CA-201**
TITLE	**Turret plate**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	24.4 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	3:4
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/turret_plate.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	4 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

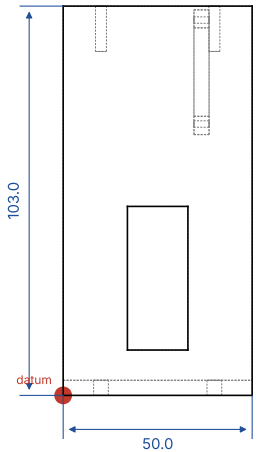
CA-202 Spine

NOTES

1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

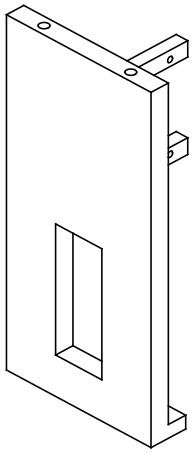




FRONT VIEW



RIGHT VIEW



ISOMETRIC (not to scale)

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	3.90	2	Z	4.0	(10.0, 8.0); (40.0, 8.0)	#6-32 clearance / running
B	2.90	2	Z	12.0	(10.0, 3.0); (40.0, 3.0)	#6-32 pilot (self-threading)
C	1.70	2	X	4.0	(18.0, 71.8); (18.0, 99.3)	servo-bag screw pilot

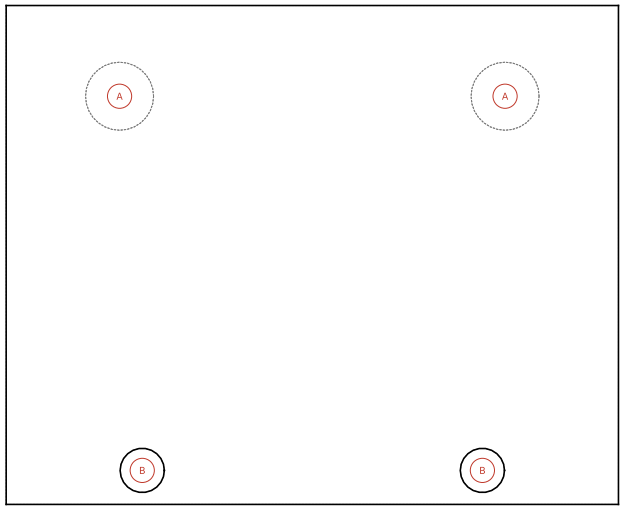
PART NO.	**CA-202**
TITLE	**Spine**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	22.4 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	1:2
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/spine.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	5 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

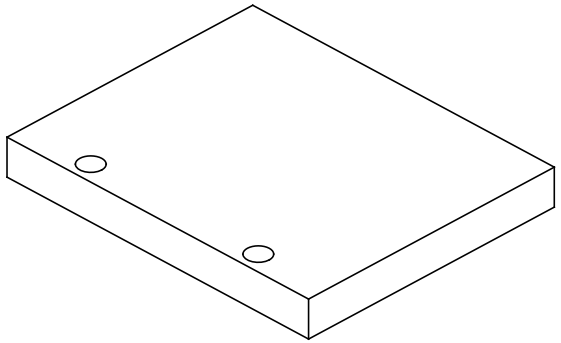
CA-203 Top cap

NOTES

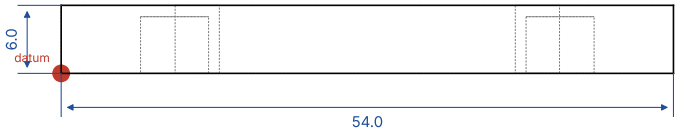
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

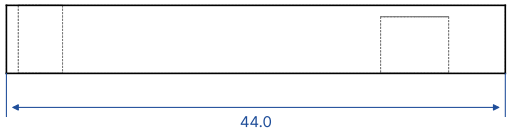




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	6.00	2	Z	5.0	(10.0, 36.0); (44.0, 36.0)	6 mm rod press fit
B	3.90	2	Z	6.0	(12.0, 3.0); (42.0, 3.0)	#6-32 clearance / running

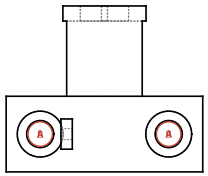
PART NO.	**CA-203**
TITLE	**Top cap**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	10.5 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	3:2
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/top_cap.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	6 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

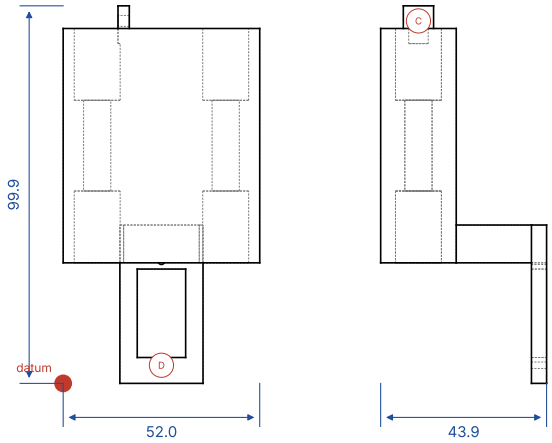
CA-301 Carriage + arm

NOTES

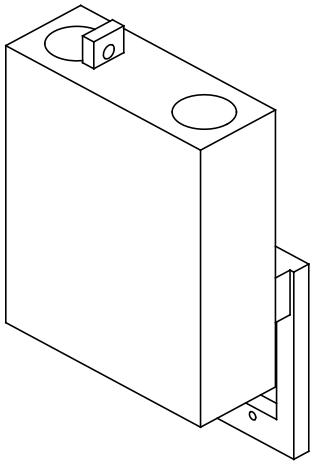
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW





FRONT VIEW RIGHT VIEW



ISOMETRIC (not to scale)

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	12.15	4	Z	19.0	(9.0, 10.0) ×2; (43.0, 10.0) ×2	LM6UU press fit
B	7.20	2	Z	24.0	(9.0, 10.0); (43.0, 10.0)	#6 pan-head pocket
C	2.90	1	X	3.0	(10.0, 95.9)	#6-32 pilot (self-threading)
D	1.70	1	Y	4.0	(26.0, 4.8)	servo-bag screw pilot

PART NO.	**CA-301**
TITLE	**Carriage + arm**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	46.0 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	1:2
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/carriage_arm.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	7 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

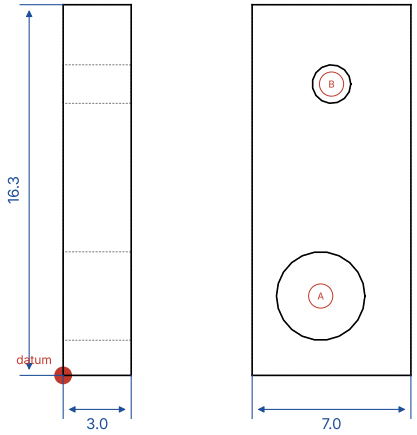
CA-302 Lift link

NOTES

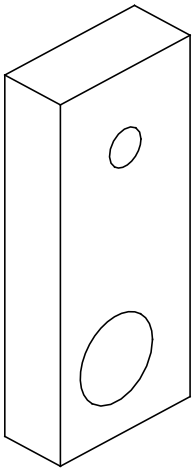
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW





FRONT VIEW RIGHT VIEW



ISOMETRIC (not to scale)

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	3.90	1	X	3.0	(3.0, 3.5)	#6-32 clearance / running
B	1.70	1	X	3.0	(3.5, 12.8)	servo-bag screw pilot

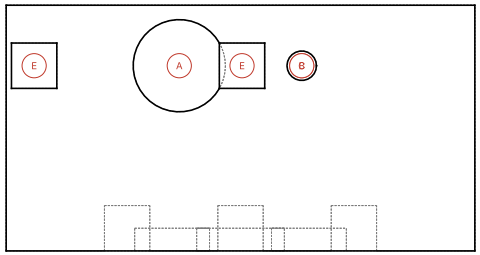
PART NO.	**CA-302**
TITLE	**Lift link**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	0.2 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	3:1
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/lift_link.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	8 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

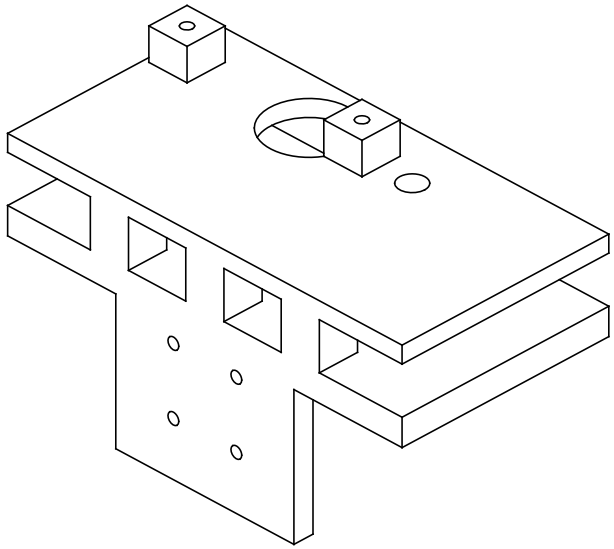
CA-401 Gripper frame

NOTES

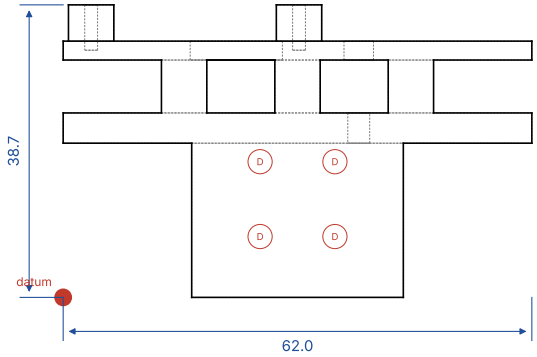
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

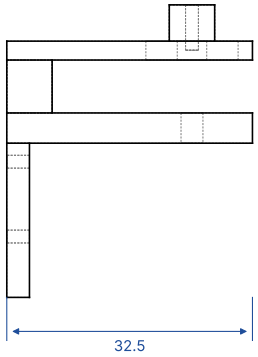




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	12.20	1	Z	2.5	(22.9, 24.5)	SG90 spline + boss clearance
B	3.90	1	Z	2.5	(39.1, 24.5)	#6-32 clearance / running
C	2.90	1	Z	4.0	(39.1, 24.5)	#6-32 pilot (self-threading)
D	1.70	4	Y	3.0	(26.1, 8.1); (26.1, 17.9) +2	servo-bag screw pilot
E	1.70	2	Z	6.0	(3.7, 24.5); (31.2, 24.5)	servo-bag screw pilot

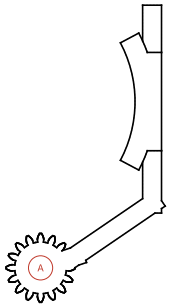
PART NO.	**CA-401**
TITLE	**Gripper frame**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	11.8 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	1:1
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/gripper_frame.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	9 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

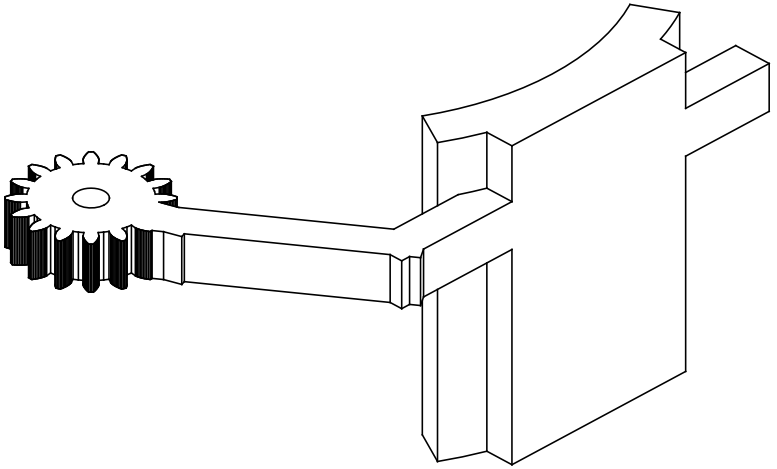
CA-402 Finger, right

NOTES

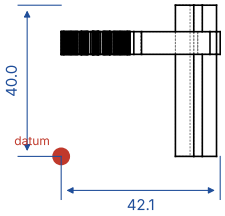
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

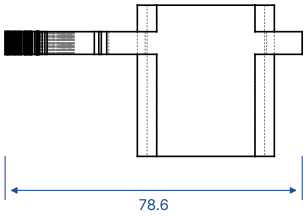




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	3.90	1	Z	6.0	(9.1, 9.1)	#6-32 clearance / running

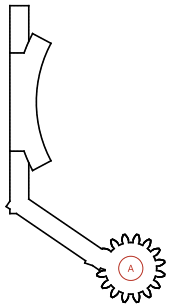
PART NO.	**CA-402**
TITLE	**Finger, right**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	9.7 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	1:2
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/finger_R.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	10 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

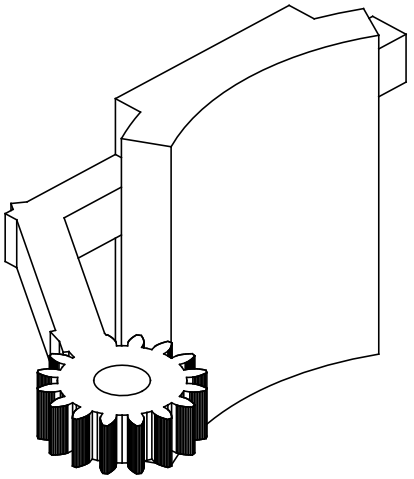
CA-403 Finger, left

NOTES

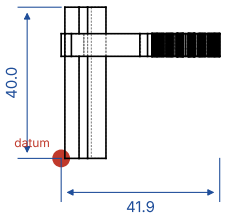
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

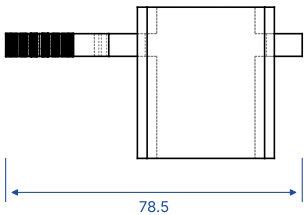




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	6.00	1	Z	4.0	(33.0, 9.0)	SG90 spline / horn socket

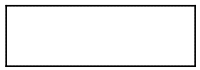
PART NO.	**CA-403**
TITLE	**Finger, left**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	9.7 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	1:2
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/finger_L.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	11 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

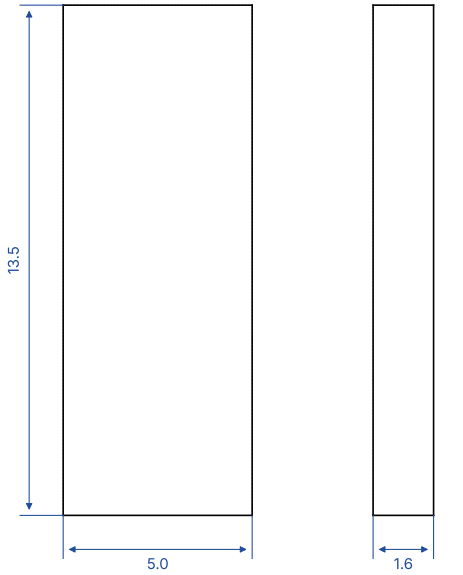
CA-404 Can bumper

NOTES

1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

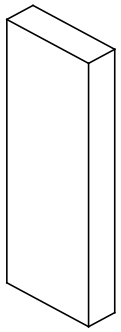
TOP VIEW





FRONT VIEW

RIGHT VIEW



ISOMETRIC (not to scale)

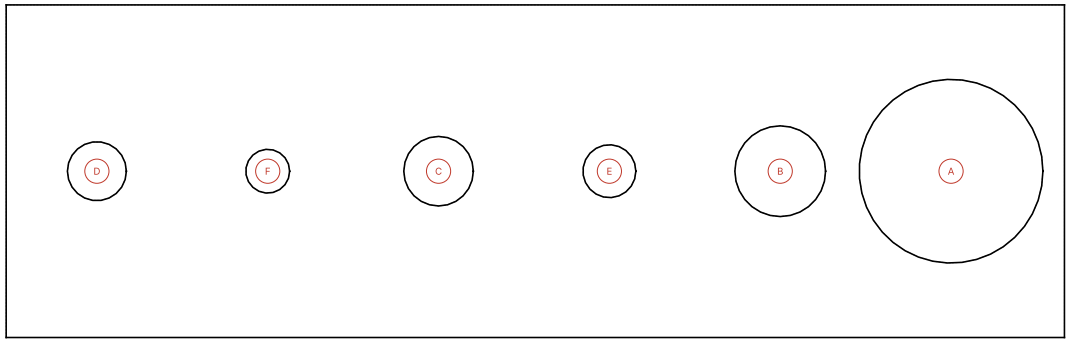
PART NO.	**CA-404**
TITLE	**Can bumper**
MATERIAL	silicone (cut from sheet)
MASS	0.1 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	5:1
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/can_bumper.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	12 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02

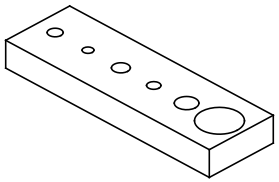
CA-900 Tolerance coupon

NOTES

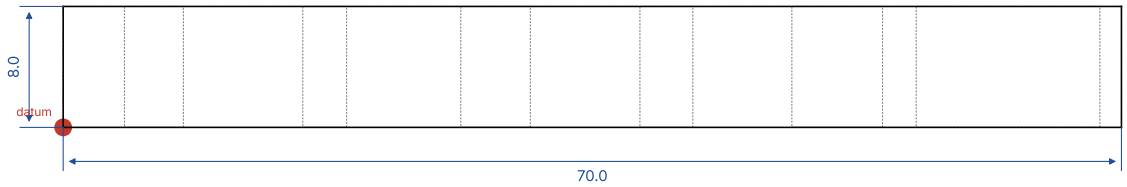
1. Hole sizes are the printed sizes from the PETG tolerance table (backend/cad/model.py TOL). Print CA-900 first and adjust TOL if a fit is off.
2. Screws thread straight into the printed pilots (HF kit #67675, no nuts).
3. Datum = the part's bounding-box corner (min X, Y, Z), lower left of the front and top views. Hole centres: (X, Y) for Z holes, (X, Z) for Y holes, (Y, Z) for X holes. ×k = k holes stacked along the axis; +n = n more positions (see the STEP file).
4. Shown in its assembly orientation; the STEP file defines all other geometry.

TOP VIEW

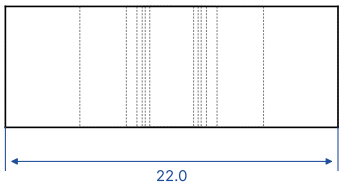




ISOMETRIC (not to scale)



FRONT VIEW



RIGHT VIEW

HOLE TABLE (letters mark hole centres on the view along that axis)

	Ø (mm)	×	axis	depth	centres from datum (mm)	fit / feature
A	12.15	1	Z	8.0	(62.5, 11.0)	LM6UU press fit
B	6.00	1	Z	8.0	(51.2, 11.0)	6 mm rod press fit
C	4.60	1	Z	8.0	(28.6, 11.0)	#8-32 clearance
D	3.90	1	Z	8.0	(6.0, 11.0)	#6-32 clearance / running
E	3.50	1	Z	8.0	(39.9, 11.0)	#8-32 pilot
F	2.90	1	Z	8.0	(17.3, 11.0)	#6-32 pilot (self-threading)

PART NO.	**CA-900**
TITLE	**Tolerance coupon**
MATERIAL	PETG printed, 0.4 mm nozzle
MASS	8.2 g (CAD, 60 % fill)
QTY	1 per robot
SCALE	2:1
UNITS	mm · general tolerance ±0.2
SOURCE	cad/step/parts/tolerance_coupon.step (from backend/cad/model.py)
PROJECTION	Third angle ⊕◁
SHEET	13 of 13
DRAWN	ME 2331 Project 2 team 2026-11-02